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Human Coronavirus 229E Infection Alters Histone Proteoforms

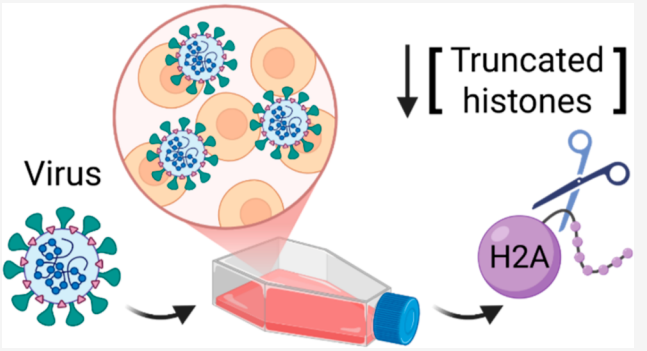
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ABSTRACT: Viruses rely on host machinery to replicate, and growing evidence demonstrates that they utilize host epigenetic regulation, including histone modification, to modulate host gene expression for their benefit. Herein, we employed top−down proteomics to quantify histone proteoforms in lung fibroblast (MRC-5) cells following human coronavirus 229E (HCoV-229E) infection and compared them to mock-infected controls. A total of 572 proteoforms were identified, including 461 histone proteoforms that were assigned to histone 2A (H2A), H2B, H3, or H4. 200 histone proteoforms were quantifiable. Differential abundance analysis revealed several changes in both reversible post-translational modifications (e.g., phosphorylation, acetylation) and the truncation states of core histones. Notably, we found a decreased abundance of truncated histones in HCoV-229E-infected samples, specifically C-terminally truncated histone H2A and N-terminally truncated histone H3. These findings underscore the power of top−down proteomics to resolve unique truncation states of proteoforms and support the hypothesis that viruses alter the histone length to influence host gene expression.



KEYWORDS: *histone, proteoforms, truncation, top−down proteomics, HCoV-229 infection*

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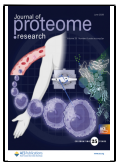
INTRODUCTION

Post-translational modifications (PTMs) of histones are critical for epigenetic regulation and allow cells to dynamically respond to their environment by mediating changes in transcription, DNA damage repair, and cell-cycle control. Acetylation, methylation, phosphorylation, and removal of PTM sites through protein truncation events are known modifications that mediate these processes by altering nucleosome dynamics and recruitment of chromatin-modifying enzymes;[1,2] these PTM events ultimately regulate transcription activation or repression.[3−5] Additionally, cleavage of the accessible termini of histones results in significant structural changes to the nucleosome subunit (primarily destabilizing the nucleosome) and removal of pre-existing PTMs.[6]

In the context of viral infections, emerging research indicates that viruses can influence the host's epigenetic mechanisms, leading to alterations in chromatin structure, which may affect the outcome of infection.[7−11] For example, herpes viruses impact the expression or activity of many histone methyltransferases (HMTs), acetyltransferases (HATs), deacetylases (HDACs), and demethylases (HDs).[12] Foot-and-mouth disease virus has been shown to N-terminally truncate histone 3 (H3) via the viral protease 3C.[13,14] Additionally, human coronaviruses (HCoV) have been demonstrated to use epigenetic mechanisms to control the expression of interferon-stimulated genes and chromatin accessibility studies demonstrated that HCoV-229E regulates host responses to infection by opening and closing chromatin access.[11,15] In MRC-5 cells specifically, HCoV-229E infection was found to alter the abundance of proteins related to mRNA splicing, translation, and ribosome biogenesis, ultimately interfering with host RNA-processing and impairing levels of host translation to enhance viral replication.[16] In addition, mass spectrometry (MS)-based assays have previously demonstrated that viral infections have measurable impacts on host histone PTM landscapes with select PTMs associated with latency or viral gene expression,[17−21] underscoring the importance of mapping host histone modifications in the context of viral infection.[22,23]

Characterizing histone proteoforms (i.e., the exact molecular form of a protein including PTMs, cleavages, single-nucleotide variants, etc.)[24] remains a challenging task.[25−27] Bottom-up strategies, in which proteins are digested to peptides, are challenged by the high number of basic amino acids in core histones, resulting in peptides that are too short for liquid chromatography (LC)−MS analysis. Additionally, characterizing unique truncation states of proteoforms is challenging via

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bottom-up approaches as semitryptic or open searches are computationally challenging and prone to incorrect assignments.[28−30] Middle-down proteomics (digestion of proteins to larger peptides >20 amino acids) has proven advantageous for histone analysis, as combinatorial PTMs can be observed on larger peptides.[31,32] However, the introduction of cleavage sites via alternative enzymes (e.g., GluC) complicates the analysis of endogenous truncation events by either making endogenous and introduced cleavages indistinguishable or complicating quantification by cutting individual proteoforms into multiple peptide species, an issue that is further exacerbated when missed cleavages occur. Top−down proteomics (protein analysis without enzymatic digestion) provides an alternative approach ideal for analyzing co-occurring PTMs or endogenous truncation events,[33,34] as the full proteoform is preserved.

Herein, we used top−down proteomics to characterize intact proteoforms of H2A, H2B, H3, and H4 enriched from human lung cells (MRC-5) following infection with human coronavirus 229E (HCoV-229E). HCoV-229E infection of MRC-5 cells is a well-established model system for investigating coronavirus−host interactions. Because MRC-5 cells retain many characteristics of normal diploid fibroblasts and HCoV-229E employs conserved strategies common to coronaviruses for establishing infection, this model provides a valuable framework for understanding how coronaviruses manipulate host signaling, metabolism, and epigenetic regulation during infection.[35,36] Overall, our top−down analysis identified a total of 461 unique histone proteoforms, of which 200 were quantifiable. We found nominally significant changes in histone processing following viral infection, including the C-terminal truncation of H2A, N-terminal truncation of H3, and changes in the abundance of reversible PTMs (e.g., acetylation, phosphorylation).

MATERIALS AND METHODS

Cell-Culture, Viral Infection, and Histone Isolation

Wild-type HCoV-229E was obtained from BEI resources (NIAID, NIH: Human Coronavirus, 229E, NR-52726). MRC-5 (human lung fibroblasts, ATCC CCL-171) cells were used to generate HCoV-229E passage 1 (p1) viral stocks. HuH7 (human liver epithelial cells, Japan Cancer Resource Bank entry JCRB0403) cells were used to generate HCoV-229E p2 viral working stocks and to quantify viral titers following infection. Viral titers were quantified using standard plaque assay approaches.[37−39] Briefly, cells (4.5×10^5 cells per well) were plated in 6-well dishes from Corning (Cat# 08-757-214). Viral samples were serially diluted, plated, and grown under 0.8% agarose and media. A 1× Neutral Red working dye was added to the plates to visualize the plaques for counting. MRC-5 cells were cultured in α-minimum essential medium (α-MEM; Gibco 12-561-056) containing 10% FBS (Cytiva) and 1% Antibiotic-Antimycotic (Gibco). HuH7 cells were cultured in low-glucose Dulbecco's MEM (Gibco 11-885-092) containing 10% FBS and 1% Antibiotic-Antimycotic. Medium serum concentrations were reduced to 4% prior to infection.

MRC-5 cells were plated, and the following day, they were either mock-infected (sterile 1× phosphate buffer saline (PBS)) or infected with HCoV-229E (multiplicity of infection MOI 3) for five replicates each. After 1 h, the inoculum was removed and the cells were washed twice with 1× PBS; fresh infection medium (4% fetal bovine serum) was added, and cultures were incubated at 34 °C at 5% CO_2 for 24 h. Mock-infected and HCoV-229E-infected cells were lysed in extraction buffer, and histones were column-enriched and acid-extracted using a Histone Purification Mini Kit according to the manufacturer's instructions (Active Motif). A bicinchoninic acid assay was then performed using BSA protein standards. Samples were then normalized to 0.1 mg/mL by diluting with 0.25% formic acid (aqueous). Samples were randomized prior to LC−MS analysis.

Top-Down Liquid Chromatography−Mass Spectrometry

Samples were analyzed using a Waters NanoACQUITY UPLC system with mobile phases consisting of 0.2% formic acid in water (mobile phase A) and 0.2% formic acid in acetonitrile (mobile phase B) as previously described.[33] Both trapping-precolumn (200 μm i.d., 5 cm length) and analytical column (100 μm i.d., 50 cm length) were slurry-packed with the C2 packing material (5 and 3 μm for trap/analytical, respectively, 300 Å, Separation Methods Technology). Samples were loaded into a 10 μL loop, corresponding to 1 μg of proteins, and injected onto the trapping column with an isocratic flow of 5% B at 5 μL/min over 15 min for desalting. After trapping and washing, samples were fractionated using the gradient parameters listed in Table S1.

The NanoACQUITY system was coupled to a Thermo Scientific Orbitrap Fusion Lumos Tribrid mass spectrometer. Source parameters included an electrospray voltage of 2.2 kV, a transfer capillary temperature of 275 °C, and an ion funnel RF amplitude of 30%. The Fusion Lumos was set to "Intact Protein" application mode, and data was collected as a full profile using previously described parameters.[40,41] MS1 data was acquired at a 120 k resolution across a 500−2000 m/z range using 4 microscans and an ACG target of 250%. A Top5 data-dependent acquisition method was used for MS2. MS2 filters included a precursor selection range from 600 to 900 m/z, an intensity threshold of 2×10^4, a charge state selection range of 12−30+, exclusion of undetermined charge states, and a dynamic exclusion of 240 s using a 0.8 m/z tolerance (excluded after $n = 1$ observation). MS2 scans were alternated between stepped higher-energy collisional dissociation (HCD) and electron capture dissociation (ETD). HCD scans were acquired using a 60 k resolution; ACG 1000%; maximum injection time of 200 ms; stepped HCD normalized collision energy (NCE) of 25, 35, and 45; 1 m/z isolation window; and 2 μscans. ETD scans were acquired using a 60 k resolution, ACG 1600%, a reaction time of 15 ms, an ETD reagent target of 7×10^5, and a 1 microscan.

Database Search for Proteoform Identification

Proteoform identification was performed with TopPIC version 1.5.4.[33,42] Settings for TopPIC included a mass error tolerance of 15 ppm for the precursor and fragment ions, a proteoform cluster error tolerance of 0.8 Da, a maximum number of two dynamic modifications (not including the N-terminal modification state), a mass shift upper bound of 500 Da, a lower bound of −500 Da, and a maximum number of one unknown mass shift. MS2 spectra were searched against the Swiss-Prot database for *Homo sapiens* containing 20,371 reviewed entries, a variably spliced ("VarSplic") database containing 21,980 splice-isoform entries, and a TrEMBL database containing 57,749 entries (UP000005640—accessed August 29th, 2021). All databases were scrambled to generate decoys that were concatenated during the search. Dynamic modifications included the following: N-terminal methionine excision; N-terminal acetylation; acetylation at lysine; C-terminal amidation; carbamylation at lysine; N-terminal carbamylation; deamidation at asparagine or glutamine; methylation at arginine or lysine; dimethylation at arginine or lysine; trimethylation at arginine or lysine; iron adduction at aspartic or glutamic acid; phosphorylation at serine, threonine, or tyrosine; ammonia neutral loss at the N-terminus or glutamine; oxidation at cysteine, methionine, or tryptophan; dioxidation at cysteine, methionine, or tryptophan; N-ethylmaleimide at cysteine; glutathionylation at cysteine. No static modifications were specified. The spectral confidence threshold in the TopPIC searches was set at a maximum allowable E-value of 0.05.

Statistical Analysis

Data analysis was performed in the R environment for statistical computing and the TopPICR package.[43] Briefly, proteoform spectrum matches were filtered to achieve a false discovery rate of 1% (FDR). A LOESS regression model was used to align retention times (RT) across datasets by comparing the RT of shared proteoforms between a reference dataset (the dataset with the most distinct proteoforms) and other datasets. These models were then used to create corrected RTs relative to the reference dataset. Proteoform masses were then recalibrated using the median ppm mass error calculated across data sets. Proteoforms were then clustered based on recalibrated mass and

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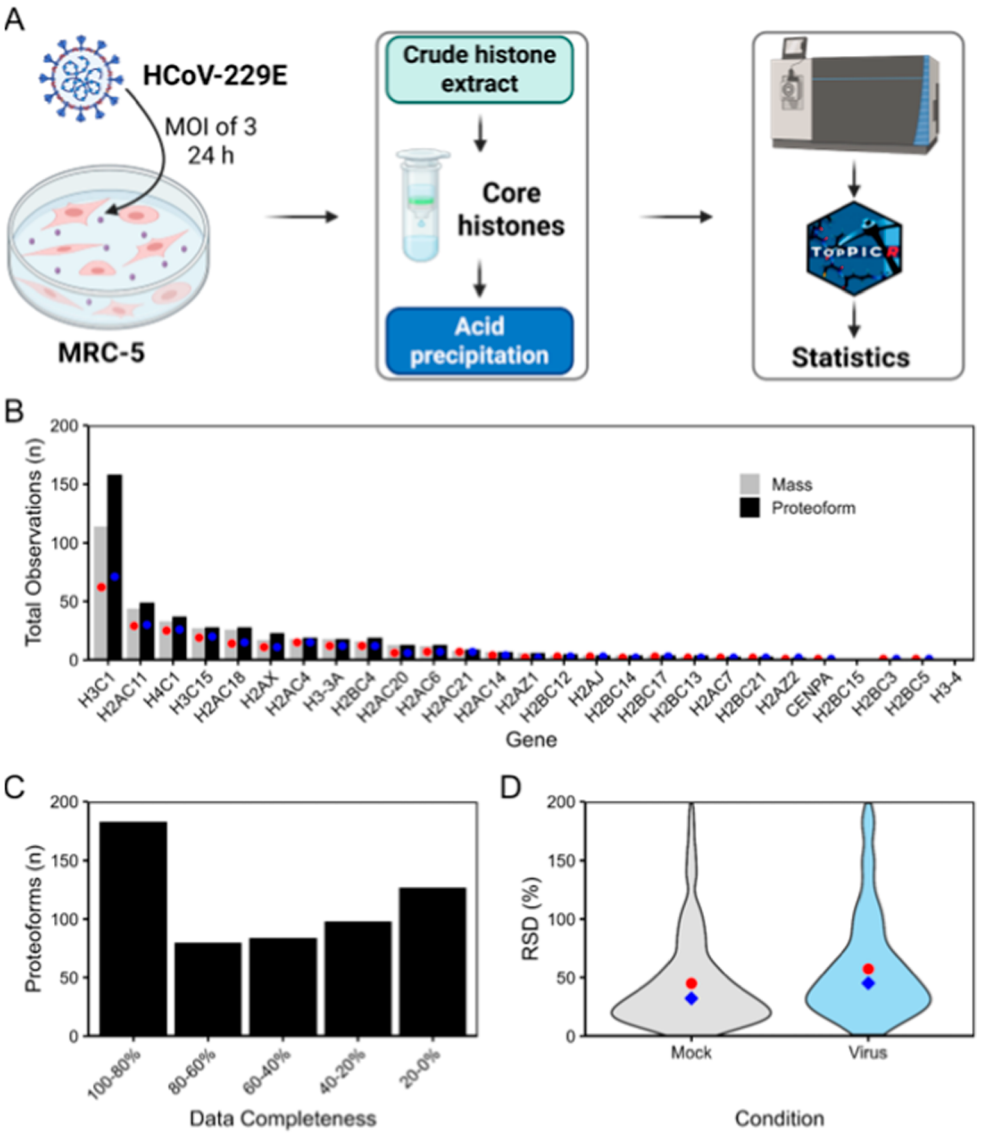


Figure 1. Overview of proteoform coverage for mock-infected or HCoV-229E-infected human lung cells. (A) Overview of experimental design for HCoV-229E-infection. (B) Total unique proteoforms (black) and monoisotopic masses (gray) per gene observed across ten datasets. Circles denote the number of observations in at least 50% of samples for proteoforms (red) and masses (blue). (C) Percentile bins containing proteoforms with different degrees of data completeness (e.g., 100% = observed in all data sets). (D) Relative standard deviations (RSD) based on label-free, intensity-based quantification of observed proteoforms. Color fill denotes either mock-infected (N = 5, gray) or HCoV-229E-infected cells (N = 5, blue). The mean RSD (red dot) and median RSD (blue diamond) are denoted.

aligned RT using hierarchical clustering. Clusters were then used to extend identifications across datasets similar to the match-between-runs (MBR) approach in bottom-up proteomics.[43] MBR was performed using the "match_features" function within the TopPICR package, which calculates a mass error (ppm) and RT tolerance window using the measured standard deviations across all datasets. Three standard deviations were allowed for the mass error and RT tolerance, resulting in cutoffs of 2.9 ppm (mass error) and 34 s (RT) for MBR assignments. After proteoform assignment, unknown mass shifts identified from the open modification search were matched to the UniMod database. An error tolerance of 0.1 Da was allowed for matching modifications and UniMod annotations. Proteoform-level and post-translational modification sites were assigned using MIScore values.[44] For quantitative analysis, data were normalized using two-way median polishing via the medpolish function in R (https://github.com/PNNL-Comp-Mass-Spec/MSnSet.utils). Proteoforms quantified in at least three replicates for both mock-infected and HCoV-229E-infected cells were used for differential analysis. Log2 fold changes were calculated using the limma R package using mock-infected samples as the ref 45. Manual MS/MS validation was performed using LcMsSpectator 1.1.7158.24217 (https://github.com/PNNL-Comp-Mass-Spec/LCMS-Spectator) using a precursor and product ion tolerance of 10 ppm, a minimum S/N threshold of 1.5, and a Pearson correlation threshold of 0.7 for fragment ion assignment. Fragmentation maps were generated using ProSight Lite v1.4 using a fragment ion tolerance of 10 ppm.[46]

RESULTS AND DISCUSSION

Top−Down Histone Characterization of HCoV-229E-Infected and Mock-Infected MRC-5 Cells

To determine how HCoV-229E infection alters the PTM status of histones, human lung fibroblasts (MRC-5 cells) were infected for 24 h, and samples were collected to assess levels of viral replication and for analysis of histones by top−down proteomics following histone enrichment using a commercial kit. As expected, HCoV-229E replicated to 10^7 plaque-forming units (pfu) per mL at 24 h post-infection, while no replication-

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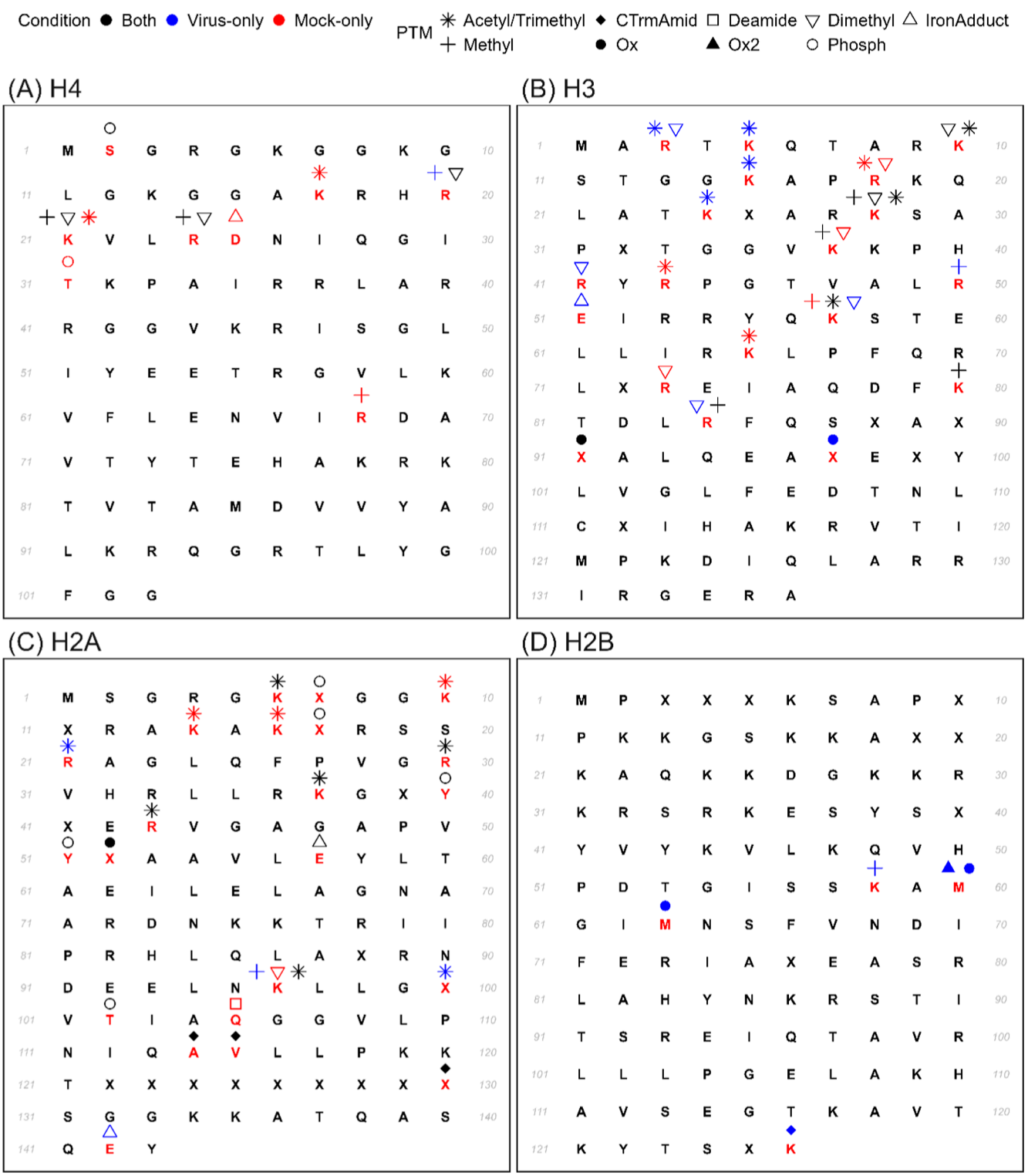


Figure 2. Localized post-translational modifications of core histones in mock- or HCoV-229E-infected MRC-5 cells. Level-1 proteoforms with localized modifications for core histones (A) H4, (B) H3, (C) H2A (excluding H2AZ and H2AV), and (D) H2B. Primary sequences are listed from top to bottom and left to right (N-terminal to C-terminal). The amino acid position is listed at the start and end of each row (italic, gray lettering). The modified amino acid is denoted with a red letter, and the modification type is shown as an overhead symbol. The symbol color denotes if the PTM is observed under HCoV-229E-infected (blue), mock-infected (red), or both conditions (black). Residues where multiple amino acids are possible due to histone variants are denoted as "X". C-terminal amidation, oxidation, or dioxidation is abbreviated as "CTrmAmid", "Ox", or "Ox2", respectively. Data are not shown for N-terminal acetylation. Note that acetylation and trimethylation cannot be distinguished at the intact mass level due to these modifications being nearly isobaric.

competent virus was detected in the mock-infected samples (Figure S1). Enriched histones from both mock-infected and HCoV-229E-infected samples were then analyzed by LC−MS. Using a 1% FDR cutoff at the gene level and selecting

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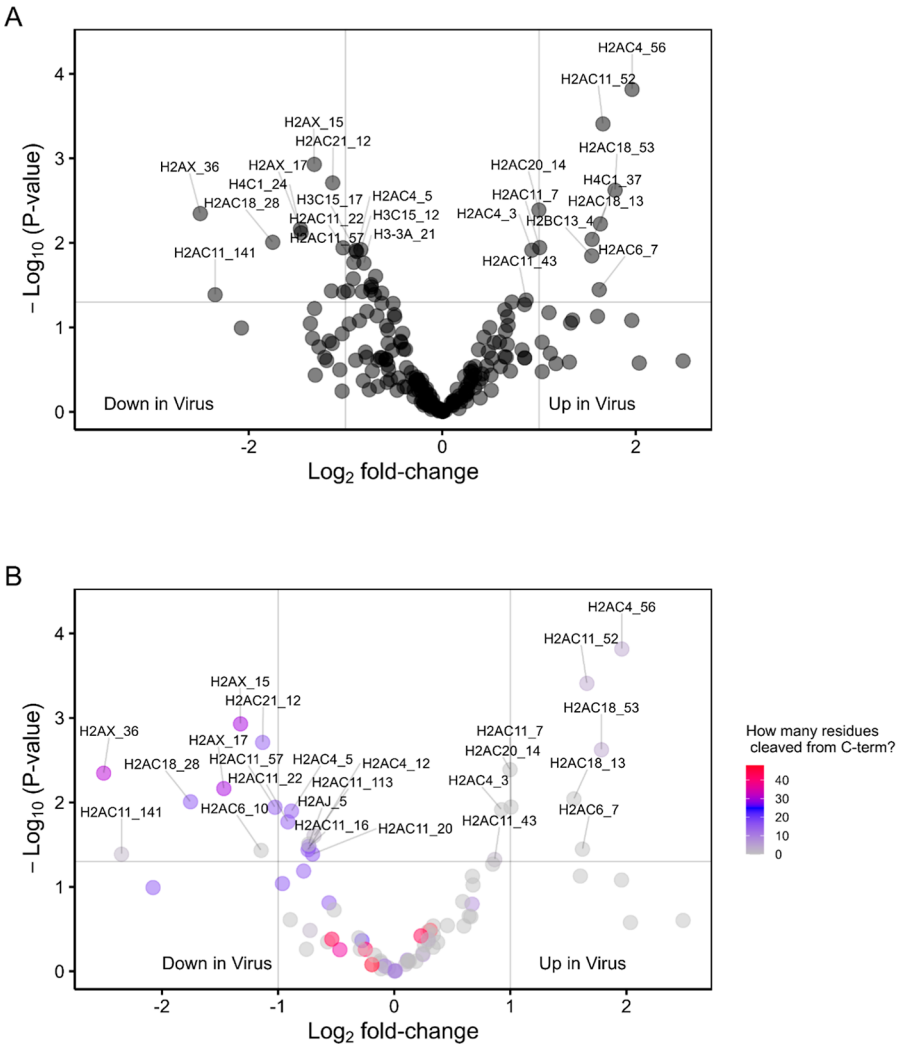


Figure 3. Differential abundance analysis of histone proteoforms following HCoV-229E infection of human lung fibroblasts. (A) Volcano plot showing all proteoforms quantified in HCoV-229E-infected MRC5 cells. (B) Volcano plot showing all H2A proteoforms quantified in HCoV-229E-infected MRC5 cells. Color fill denotes how many residues are cleaved from the canonical C-terminus. Proteoforms that are significant ($p < 0.05$) are annotated as "gene_proteoform identifier". Horizontal line indicates p-value cutoff (0.05) and vertical lines indicate log2 fold-change cutoff of 1 and −1.

proteoforms with at least two spectral matches, we identified a total of 572 proteoforms across 10 datasets from mock-infected and HCoV-229E-infected human lung MRC-5 cells (Supporting Information S1); this included 461 histone proteoforms that were assigned to H2A, H2B, H3, or H4. Because many PTMs cannot be fully localized (level 2A proteoforms),[47] we define proteoforms as species that share the same monoisotopic mass, gene, N-terminus, C-terminus, and elute within 3 standard deviations of the centroid elution time for a given proteoform; thus, a proteoform may encompass several isometric PTM species.

We observed numerous proteoforms per histone gene, including H3 (H3C1, H3C15, H3-3A, H3-4); H4 (H4C1); H2A (H2AC11, H2AC18, H2AX, H2AC4, H2AC6, H2AC21, H2AC14, H2ACZ1, H2AC7, H2AJ, H2ACZ2); H2B (H2BC4, H2BC13, H2BC12, H2BC17, H2BC21, H2BC14, H2BC15, H2BC3, H2BC5); and centromere protein A (CENPA). Of these isoforms, H3C1 was unsurprisingly the most complex in terms of unique proteoforms and masses ($n = 158$, $n = 114$), followed by H2AC11 ($n = 49$, $n = 44$), and H4C1 ($n = 37$, $n = 33$) (Figure 1A). We observed a high degree of data completeness (i.e., the number of samples a proteoform is observed in/the total number of samples) using the match-between-runs approach, with 263 proteoforms (46%) observed in at least 6 out of 10 samples (Figure 1B). Of the total 3324 proteoform assignments, 871 matches were MBR-based (data not shown). Additionally, the distribution of relative standard deviations (RSDs) of label-free intensities was within the expected range of technical variation based on previous top–down label-free quantitation (LFQ) studies (Figure 1C).[33,34] Moreover, RSDs were uniform across the intensity range for quantifiable proteoforms (proteoforms observed in at least three of the five biological replicates for HCoV-229E-infected and mock-infected samples; Figure S2).

The PTM Landscape of Core Histones in Mock-Infected and HCoV-229E-Infected MRC-5 Cells

To establish a comprehensive view of the histones in mock- or HCoV-229E-infected MRC-5 cells, we next mapped the PTM landscape of the core histones using sites that can be fully localized (level 1 proteoforms).[47] Our analyses revealed a diverse repertoire of methylation, phosphorylation, and acetylation marks across histones H4, H3, H2A, and H2B (Figure 2). Select proteoforms are observed in only HCoV-

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229E-infected or mock-infected samples; however, it is unclear if these results are statistically significant given the potential for missing values in data-dependent acquisitions. All reported PTM positions were based on alignments summarized in Supporting Information S2, assuming the leader methionine as position 1. For H4, we observed phosphorylation of S2 and T31; methylation of R20, K21, R24, and R68; dimethylation of R20, K21, and R24; and acetylation or trimethylation at K17 and K21 (Figure 2A). Because acetylation and trimethylation are nearly isobaric and cannot be distinguished at the intact mass level, these modifications were listed ambiguously. In addition, partially localized modifications of H4 (Table S2) were also observed, which were assigned to all possible amino acid residues. For example, methylation can be partially localized to multiple lysine and arginine residues between K17 and R40.

For H3, we observed methylation at K28, K37, R50, K57, K80, and R84; dimethylation at R3, K10, R18, K28, K37, R41, K57, R73, and R84; and acetylation/trimethylation at R3, K5, K10, K15, R18, K24, K28, R43, K57, and K65 (Figure 2B). Partially localized modifications of H3 are provided in Table S3. These data revealed additional N-terminal modifications, with methylation and acetylation localized between R3 and K24 (encompassing eight possible R/K sites) and glutathionylation of H3C1 localized to C97 and/or C111. Prior reports have localized this glutathionylation to C111 and demonstrated that it destabilizes the nucleosome complex.[1,48]

For H2A (excluding H2AZ and H2AV), we observed phosphorylation at T7, S17, Y40, Y51, and T102; C-terminal amidation of various truncated forms at A114, V115, and X130; methylation and dimethylation at K96; and acetylation/trimethylation at numerous sites, including K6, K10, K14, K16, R21, R30, R37, R43, K96, and K100 (Figure 2C). While we cannot unambiguously assign acetylation versus trimethylation, these H2A modifications are likely acetylation based on prior reports.[49] Partially localized PTMs on H2A included several methylations, dimethylations, and trimethylations/acetylations (Table S4). H2AZ and H2AV are listed separately, given their low sequence homology. PTMs are limited and include various deamidations, C-terminal amidation of the canonical C-terminus (Figure S3), and acetylation/trimethylation at R40 and R46 (Table S5). Finally, for H2B, we characterized methylation at K58, various oxidations, and C-terminal amidation (Figure 2D).

HCoV-229E Infection Reduces Truncated Histone Accumulation

We next investigated the differential proteoform abundance between mock-infected and HCoV-229E-infected samples. Proteoforms observed in at least 3 replicates per condition were considered, leaving 200 quantifiable histone proteoforms. We opted to use unadjusted *p*-values given the small number of statistical tests performed in comparison to other shotgun proteomics studies with thousands of quantifications, where the number of potential Type I errors is higher. Additionally, we emphasize an absolute log2 fold-change cutoff >1, given the median CVs of 32% and 45% (Figure 1D, for mock and virus samples, respectively); this filter ensures the observed changes in proteoform abundance exceed the variability of our measurements.

Applying an unadjusted *p*-value cutoff of <0.05 and log2 fold-change cutoff of less than −1 or greater than 1, we found 18 differential histone proteoforms in HCoV-229E-infected samples in comparison to mock-infected samples (Figure 3A, Supporting Information S3). Many of these species were H2A proteoforms, with C-terminally truncated species (>15 amino acids truncated) being reduced in HCoV-229E-infected samples and full-length H2A proteoforms being more abundant in HCoV-229E-infected samples relative to mock (Figures 3B and S4). The H2A proteoforms with the lowest assigned p-values were cleaved from C-terminal to amino acids 113 or 114 (LPNIQ.A or LPNIQA.V, where "." denotes cleavage), with these proteoforms being less abundant in HCoV-229E-infected samples versus mock-infected controls (Table S6). By contrast, H2A proteoforms with increased abundance in HCoV-229E-infected cells were full-length or cleaved at position 126 (ESHHK.A) (Table S6). Reversible PTMs (e.g., acetylation/trimethylation and phosphorylation) did not appear to influence the fold-change direction, which was instead associated with the truncation state (Figure S4). C-terminal truncation of H2A near amino acids 113/114 has previously been reported from bottom-up studies via semitryptic peptides (LPNIQAV.L), including in human-derived B cells.[50] Additionally, extensive C-terminal cleavage of H2A and H2B has been observed via top−down proteomics in butyrate-treated HeLa cells.[51]

We also observed a consistent decrease in the abundance of N-terminally truncated forms of H3, specifically forms cleaved from the N-terminal to R27 and K28 (Figures S5 and S6) following HCoV-229E infection. Of the nine quantifiable proteoforms cleaved at these sites, eight had decreased differential expression, with three being nominally significant (*p*-value <0.05). For these species, reversible PTMs did not appear to influence the fold-change direction (Figure S6). Truncation of residues 1−27 of H3 has also been shown to decrease nucleosome stability.[52] Of note, the foot-and-mouth disease virus has been shown to N-terminally truncate histone H3 via protease 3C;[13,14] however, the exact cut site remains unknown and would be predicted to enhance rather than restrict levels of host transcription. The H3 cut sites (R27 and K28) have previously been observed in mouse embryonic stem cells and human peripheral blood mononuclear cells.[53,54]

Overall, our data reveal decreased abundance of truncated histones H2A and H3 following HCoV-229E infection. Several histone-specific proteases have been previously described that perform this "clipping" function, including two H2A-specific proteases[55−58] and several multifunctional proteases that cleave H3.[13,14,53,54] The consequences of histone truncation on the nucleosome stability have been explored in other model systems unrelated to viral infection. Studies in HEK293T cells using fusion constructs of full-length and C-terminally truncated H2A have demonstrated that C-terminal cleavage has several consequences including lowering nucleosome stability, disrupting recruitment of linker histone H1, and increasing cell sensitivity to nongenotoxic stress (e.g., osmotic stress).[59] It is currently unclear how this H2A cleavage alters gene expression inside cells; however, a decrease in nucleosome stability could potentially increase DNA accessibility and gene expression.

We hypothesize that the decreased abundance of truncated histones could result from virally encoded histone mimics that negatively regulate the functions of histone proteases to reduce the level of transcription following infection. For example, H3N2 strains of the influenza virus and SARS-CoV 2 encode nonstructural proteins (e.g., SARS-CoV 2 open reading frame 8) that function as H3 mimics and contribute to alterations of host transcription to enhance the intracellular environment for replication.[60] HCoV-229E nonstructural proteins do not share sequence homology with SARS-CoV 2's ORF8, but there may

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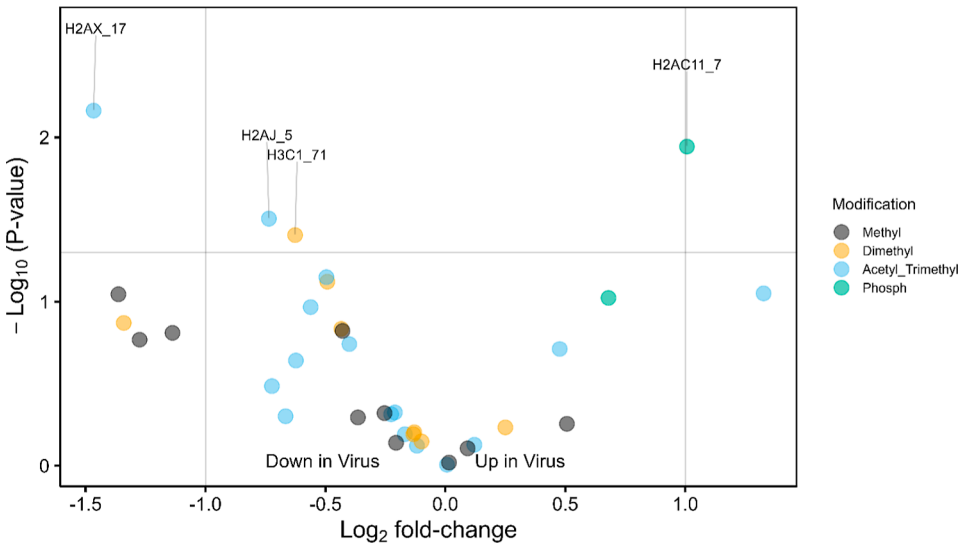


Figure 4. Differential abundance analysis of methylated and acetylated histone proteoforms in HCoV-229E-infected human lung fibroblasts. Volcano plots of methylated and acetylated proteoforms were identified following HCoV_229E infection. Color fill denotes methylation (mono = black; di = gold) or acetylation and trimethylation (blue). Proteoforms are annotated as "gene_proteoform identifier". The horizontal line indicates the *p*-value cutoff (0.05), and vertical lines indicate the log2 fold-change cutoff of 1 and −1.

be a functional domain within an HCoV-229E protein that could have a similar histone-mimicking function.[61]

Regulation of Histone Proteoforms with Reversible PTMs by HCoV-229E Infection

Changes in reversible PTMs were observed, including increased phosphorylation of H2AC11-T17 in HCoV-229E-infected samples (Figure 4). Phosphorylation of H2A-T17 has been previously reported as a marker of oxidative stress *in vitro*,[62] and HCoV-229E infection has been shown to cause oxidative stress in MRC-5 cells.[63] Thus, it is intriguing to hypothesize that virus-induced oxidative stress results in increased levels of phosphorylated H2A-T17 in our samples, giving the broad crosstalk between redox- and phosphorylation-based signaling. We also observed decreased acetylation/trimethylation of H2AX-K6 in the HCoV-229E-infected samples. Previous studies demonstrated that H2AX-K6 is acetylated in response to DNA double-strand breaks, and ultimately regulates acetylation-dependent ubiquitination of H2AX, H2AX release from chromatin, and chromatin reorganization following DNA damage.[64,65] Finally, we observe decreased dimethylation of N-terminally truncated H3C1−K28 in HCoV-229E-infected samples; however, this proteoform is N-terminally cleaved at K28 and it is likely that the truncation drives differential abundance as described above. Fragmentation maps for the discussed PTMs are provided as Figures S7−S10.

Overall, we observed significant changes in reversible PTMs following HCoV-229E infection. However, our data were constrained by limited proteoform coverage due to the inherent limitations of data-dependent acquisition methods and insufficient fragmentation coverage, which restricted the ability to quantify isomeric modification states based on assigned fragment ions.[66] Given these limitations, we also utilized pyMS-Vis,[67] an MS1-based quantification tool, to test for HCoV-229E-induced changes in deconvoluted masses in the expected histone molecular weight (10−25 kDa). This approach returned the same measured changes for H2A proteoforms as identified by TopPIC but not for other differentially abundant masses (not shown).

This study has several limitations including limited sample size, limited time course data, and, importantly, limited PTM localization that restricts our quantification to level 2 proteoforms. Additionally, several modification types, including cysteine oxidations, are difficult to characterize without specific PTM profiling workflows.[68,69] Despite these limits, our findings underscore the power of top−down proteomics to resolve unique truncation states of proteoforms and support the hypothesis that viruses alter the histone length to influence host gene expression. Top−down proteomics is a rapidly expanding field, and we anticipate that instrumentation and method developments will improve future histone characterization without relying on complex, 2D, or offline fractionation.[70] In particular, the expansion of data-independent acquisitions into top−down could greatly alleviate the failed characterization of lower-intensity precursor ions.[71,72]

CONCLUSIONS

We used top−down proteomics to characterize histone proteoforms from human lung cells following HCoV-229E infection. We identified 461 unique histone proteoforms and quantified 200 histone proteoforms, uncovering changes in core histone truncations and reversible post-translational modifications. Notably, the decreased abundance of C-terminally truncated H2A and N-terminally truncated H3 suggested potential mechanisms by which the virus modulates nucleosome dynamics and, subsequently, alters host gene expression. These findings align with prior studies reporting epigenetic changes induced by HCoV-229E infection and support the potential for truncation events to serve critical roles in transcriptional regulation during the host−virus interactions. This study highlights the advantages of top−down proteomics in uncovering intact proteoforms and generating deeper insights into histone modifications, particularly protease-driven truncation events that cannot be fully resolved by bottom-up approaches. Further studies should reveal the dynamic PTM codes in response to viral infection, and the information gained can be used to provide a rational foundation for designing host-directed antiviral therapies that modulate chromatin regulation.

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■ ASSOCIATED CONTENT

Data Availability Statement

The mass spectrometry raw data have been deposited to the ProteomeXchange Consortium via the MassIVE partner repository with data set accession MSV000099293. Code AvailabilityScripts for TopPICR analysis, differential abundance analysis, and figure generation are publicly available on GitHub at https://github.com/ashleyives/top_down_viral_histones.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.jproteome.5c01015.

Figure S1: HCoV-229E replication levels in MRC-5 Cells; Figure S2: relative standard deviation (RSD) versus log10 median label-free intensity for quantifiable histone proteoforms in mock- and HCoV-229E-infected human lung cells; Figure S3: localized post-translational modifications of H2AZ and H2AV in mock- and HCoV-229E-infected human lung cells; Figure S4: truncation state and post-translational modifications of quantifiable H2A proteoforms in mock- and HCoV-229E-infected human lung cells; Figure S5: differential abundance of N-terminally truncated H3 proteoforms in HCoV-229E-infected human lung cells; Figure S6: truncation state and post-translational modifications of quantifiable H3 proteoforms in mock- and HCoV-229E-infected human lung cells; Figure S7: fragmentation map of acetylated/trimethylated H2AJ in mock- and HCoV-229E-infected human lung cells; Figure S8: fragmentation map of acetylated/trimethylated H2AX in mock- and HCoV-229E-infected human lung cells; Figure S9: fragmentation map of phosphorylated H2AC11 in mock- and HCoV-229E-infected human lung cells; Figure S10: fragmentation map of dimethylated H3C1 in mock- and HCoV-229E-infected human lung cells; Table S1: liquid chromatography gradient used for separation of intact histones from mock- and HCoV-229E-infected human lung cells; Table S2: partially localized modifications of H4 in mock- and HCoV-229E-infected human lung cells; Table S3: partially localized modifications of H3 in mock- and HCoV-229E-infected human lung cells; Table S4: partially localized modifications of H2A (excluding H2AV and H2AZ) in mock- and HCoV-229E-infected human lung cells; Table S5: partially localized modifications of H2AV and H2AZ in mock- and HCoV-229E-infected human lung cells; Table S6: C-terminal cleavage sites for differentially abundant H2A proteoforms in mock- and HCoV-229E-infected human lung cells (PDF)

All identified proteoforms in this study, including gene, primary sequence, post-translational modifications, aligned retention time, and monoisotopic mass (XLSX)

Histone sequence alignments used to describe the PTM position (XLSX)

Differential abundance data for quantifiable histone proteoforms (XLSX)

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Notes

This research did not involve human or animal participants. The authors declare no competing financial interest.

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■ ABBREVIATIONS

AGC	automatic gain control
ETD	electron-transfer dissociation
FA	formic acid
FDR	false discovery rate
HCD	higher-energy collisional dissociation

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HCoV-229E	human coronavirus 229E
LC-MS/MS	liquid chromatography−tandem mass spectrometry
LFQ	label-free quantification
MBR	match-between-runs
MOI	multiplicity of infection
MS	mass spectrometry
NCE	normalized collision energy
TDP	top−down proteomics

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